Session 1 : Cloud Computing Fundamentals

Agenda for Today's Session

1. Cloud Computing Basics: Definition & the NIST view; on-premise IT vs cloud; drivers: cost, scale & speed
2. History & Evolution: Mainframes & time-sharing; client-server & virtualization; grid & utility computing; the public cloud era
3. Essential Characteristics - I: On-demand self-service; broad network access; resource pooling & multi-tenancy
4. Essential Characteristics - II: Rapid elasticity vs scalability; measured service & pay-per-use metering
5. Benefits & Challenges: Cost model shift, elasticity, global reach & speed to market; vendor lock-in, cost overruns, latency & data residency

---------- SECTION 1: 1. Cloud Computing Basics ----------

Slide 1: Why Cloud Computing Emerged

Heading: Cloud Computing Overview

Subheading: Why Cloud Emerged

Content:

Traditional IT ties hardware purchases to guesses about future demand, locking capital years before the load actually arrives.

- On-premise IT: organisation buys, installs, maintains hardware
- Cloud IT: computing rented as a metered service
- Shift driven by cost, scale and speed pressures

Visual Guidance: Split-screen diagram: left a server room with racks and IT staff, right a laptop reaching a cloud icon over the network.

Speaker Notes: "Set up the capacity-planning problem before naming cloud computing as the fix. Interviewers often open this topic by asking what problem cloud computing actually solves."

Slide 2: Defining Cloud Computing Formally

Heading: Cloud Computing Defined

Subheading: NIST Formal View

Content:

The National Institute of Standards and Technology (NIST) defines cloud computing as a model for enabling anywhere, convenient, on-demand network access to a shared pool of configurable computing resources (networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

- Five essential characteristics govern shared resource access .
- Three service models: IaaS, PaaS, SaaS .
- Four deployment models: public, private, hybrid, community .
- AWS Management Console exposes this self-service resource catalog .

Analogy: Renting a furnished apartment instead of constructing a house gives instant living space without owning bricks or cement — just as cloud computing gives access to configurable computing resources on demand without owning the underlying infrastructure.

Visual Guidance: Central box labelled 'NIST Definition' branching into three labelled boxes: Essential Characteristics, Service Models, Deployment Models.

Speaker Notes: "Name NIST SP 800-145 as the source so the definition is quoted correctly. This exact definition is a frequent interview question, so state it precisely."

Slide 3: On-Premise IT vs Cloud

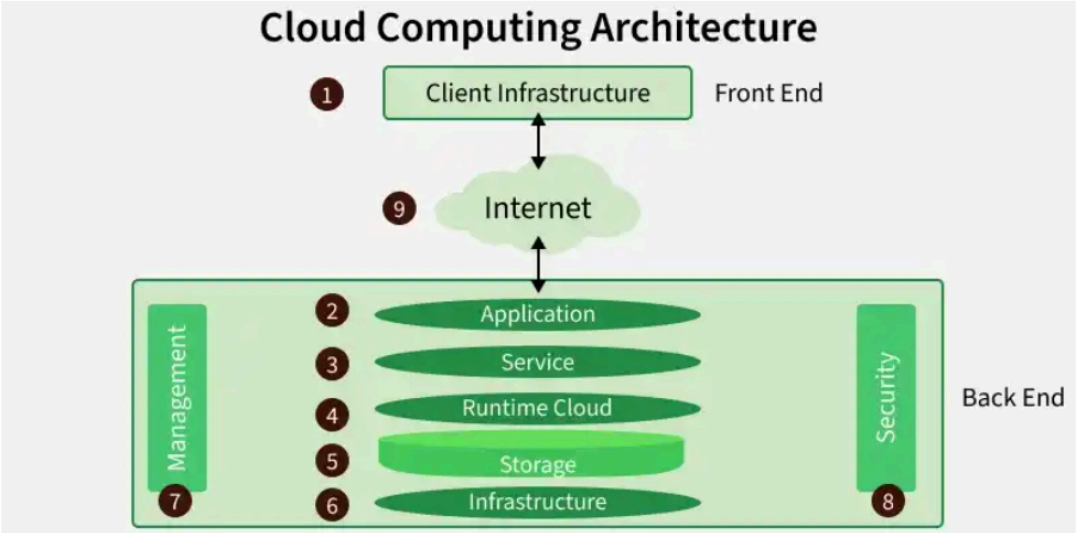
Heading: On-Premise vs Cloud

Subheading: Ownership and Delivery

Content:

On-premise IT keeps hardware, software and staff inside the organisation; cloud shifts ownership and operation to a provider, delivered over the network.

Aspect	On-Premise IT	Cloud
Ownership	Organisation owns hardware	Provider owns infrastructure
Cost model	Capital expenditure (CapEx)	Operating expenditure (OpEx)
Provisioning time	Weeks to months	Minutes
Scaling	Manual hardware upgrade	On-demand, software-driven
Maintenance	In-house IT team	Provider-managed



Visual Guidance: Two-panel diagram: left a locked data-centre building, right a cloud icon reachable from anywhere over the internet.

Speaker Notes: "Stress the CapEx-to-OpEx shift as the single biggest contrast for interviews. Ask learners to map each row to a real purchase decision they know."

Slide 4: Cost Scale and Speed Drivers

Heading: Adoption Drivers

Subheading: Cost Scale Speed

Content:

Three pressures push organisations toward cloud: unpredictable cost of owning hardware, difficulty scaling for sudden demand, and slow speed of provisioning new infrastructure.

- Cost: Pay for what you use instead of maintaining unused hardware.
- pEx to pay-per-use OpEx, cuts idle capacity
- Scale: resources added or removed as demand shifts
- Speed: servers provisioned in minutes, not procurement cycles
- Reflected in tools like AWS Cost Explorer, Pricing Calculator

Visual Guidance: Three icons in a row — a rupee coin, an elastic band, a stopwatch — each labelled cost, scale, speed, feeding one arrow marked 'Cloud Adoption'.

Speaker Notes: "Tie each driver to a concrete AWS tool so the point feels practical, not abstract. Interviewers often ask learners to justify a cloud migration using exactly these three drivers."

---------- SECTION 2: 2. History & Evolution ----------

Slide 5: Four Eras of Computing Evolution

Heading: Four Eras Of Computing

Subheading: From Mainframes To Cloud

Content:

Cloud computing did not appear overnight; it inherited ideas from four earlier eras, each solving the resource-sharing problem of its own time.

- 1960s-70s: mainframes & time-sharing
- 1980s-2000s: client-server & x86 virtualization
- 1990s-2000s: grid & utility computing
- 2006 onward: the public cloud era

Visual Guidance: Horizontal timeline strip with four labelled eras and simple icons: mainframe terminal, PC-plus-server, linked grid nodes, cloud symbol.

Speaker Notes: "Frame this as the roadmap for the section before going era by era. Exam angle: each era is often asked as 'what did X contribute to cloud computing'."

Slide 6: Mainframes and the Rise of Time-Sharing

Heading: Mainframes & Time-Sharing

Subheading: One Machine Many Users

Content:

Early mainframes were too costly for one user alone, so time-sharing let many users access the same machine at once.

- Mainframes: centralised, costly machines run by dedicated operators
- Batch processing: jobs queued and run without interaction
- Time-sharing (1960s): CPU scheduled across simultaneous terminal users
- Each user experienced an illusion of a private computer

- IBM's CP-40/CP-67 (1960s) pioneered mainframe virtualization — the earliest form of the concept

Analogy: A library reading room with one shared encyclopedia set, where each reader gets a scheduled turn at the desk — just as time-sharing gave every terminal user a scheduled turn on one mainframe.

Visual Guidance: Mainframe box with several terminal lines converging on it, a small clock icon marking scheduled turns.

Speaker Notes: "Time-sharing is the first appearance of the resource-multiplexing idea cloud scales up massively. Exam angle: contrast queued batch jobs against interactive time-sharing."

Slide 7: Client-Server Shift and Virtualization

Heading: Client-Server & Virtualization

Subheading: Hardware Gets Abstracted

Content:

Personal computers and networks split work into client requests and dedicated servers; virtualization then let one physical server run several independent virtual machines — self-contained, software-based "computers" that each run their own OS .

- 1980s-90s: client-server split desktop and network-server roles
- Commodity x86 virtualization (late 1990s onward, e.g. VMware): a hypervisor — the software layer that creates and manages VMs — partitions one server
- Each virtual machine runs its own OS, isolated from others
- Decouples software from physical hardware, enabling resource pooling

1980s–90s: Client computers sent requests to dedicated servers.

Clients: Computers used by people to access applications and services.

Servers: Computers designed to provide data, applications, or services to other computers.

This allowed many clients to use shared resources from dedicated servers.

Next: As the number of servers grew, a new idea emerged — **virtualization**, which helped use physical servers more efficiently.

Analogy: One large plot subdivided into separate, locked apartments by a building's structure — just as a hypervisor partitions one physical server into isolated virtual machines, each unaware of its neighbours.

Visual Guidance: Diagram: physical server box, hypervisor layer above it, three VM boxes each with its own OS icon on top.

Speaker Notes: "Virtualization is the direct technical ancestor of cloud multi-tenancy taught later in this session. Interview angle: explain how a hypervisor isolates VMs sharing one host."

Slide 8: Grid Computing and Utility Computing

Heading: Grid & Utility Computing

Subheading: Pooling And Metering Power

Content:

Grid computing pooled processing power across many organisations for one large task, while utility computing proposed billing that powers the way electricity is billed.

- Grid computing:Combines unused processing power from many computers for large tasks.
- Example : Scientific projects use processing power from many participating computers.
- Utility computing: Pay for the computing capacity you actually use, rather than owning it.
- Key idea: Both concepts separate resource usage from hardware ownership, paving the way for cloud computing.

Analogy: Several small businesses sharing one shipping company's trucks, pooling partial loads and paying only for the weight each one ships — just as grid computing pools spare processing power and utility computing bills only for capacity consumed.

Visual Guidance: Diagram: multiple institution icons linked into one central grid; a separate metering gauge icon beside it for billing.

Speaker Notes: "Grid pioneered pooling distributed compute; utility pioneered pay-per-use billing, both feeding directly into cloud. Interview angle: distinguish grid (pooled computation) from cloud (pooled plus on-demand self-service)."

Slide 9: The Public Cloud Era Begins

Heading: Public Cloud Emerges

Subheading: 2006 Onward Milestones

Content:

The public cloud era began when providers started selling the pooled, virtualized, metered resources of earlier eras directly over the internet, on demand.

2006: Amazon Web Services launches S3 storage, then EC2 compute
2008: Google App Engine offers a managed application platform
2010: Microsoft launches Windows Azure commercially

- By the 2010s: public cloud becomes the default for new IT

Visual Guidance: Timeline strip from 2006 to 2010s with provider logos placed at their launch years and a rising adoption curve beneath.

Speaker Notes: "AWS's 2006 launch is the commonly cited start of the modern public cloud industry. Exam angle: know that S3 preceded EC2 within that same year."

Slide 10: Evolution Timeline and Legacy Summary

Heading: Evolution At A Glance

Subheading: Eras And Their Legacy

Content:

Each era solved a resource problem that cloud computing still answers, from shared access on one machine to metered, on-demand delivery over the internet.

Era	Period	Core Idea	Cloud Legacy
Mainframes & Time-sharing	1960s-70s	Many users share one machine	Shared-access roots
Client-Server & Virtualization	1980s-2000s	Hardware abstracted into virtual machines	Resource pooling
Grid & Utility Computing	1990s-2000s	Pooled capacity, pay-per-use billing	Metered service model
Public Cloud	2006 onward	On-demand delivery over internet	Today's cloud platforms

Visual Guidance: Full four-row table on screen with a slim timeline ribbon above it repeating the four eras in sequence.

Speaker Notes: "This table is a frequent interview question on cloud history. Ask learners to match each precursor model to the cloud characteristic it most resembles."

---------- SECTION 3: 3. Essential Characteristics - I ----------

Slide 11: NIST Essential Characteristics

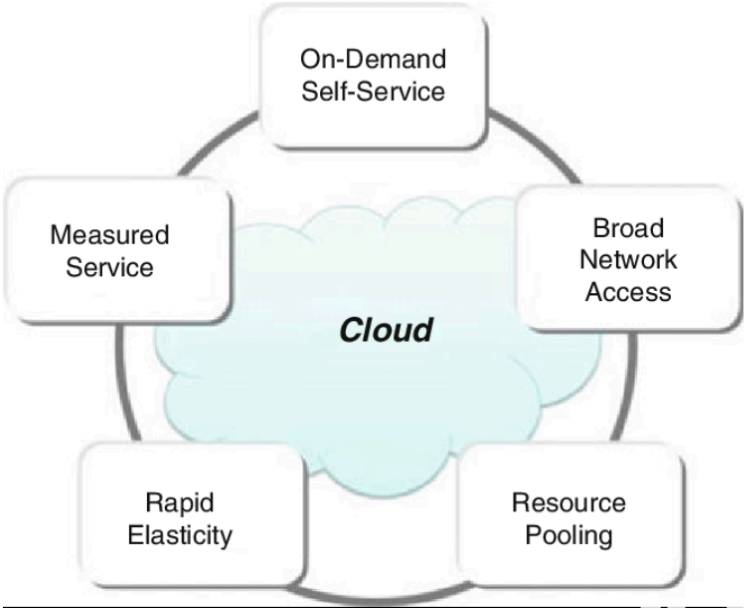
Heading: Essential Characteristics

Subheading: Five NIST Traits

Content:

NIST formally defines cloud computing through five essential characteristics, the standard checklist that distinguishes cloud from traditional IT.

- On-demand self-service: provisioning without human intervention
- Broad network access: reachable over standard protocols
- Resource pooling: shared infrastructure across multiple tenants
- Rapid elasticity: capacity scales automatically with demand
- Measured service: usage metered and billed per consumption



Visual Guidance: Five-box diagram, each box labelled with one NIST characteristic name, arranged around a central 'Cloud' icon.

Speaker Notes: "The five traits together form the complete NIST definition of cloud computing. The next three slides explain on-demand self-service, network access and pooling in more depth."

Slide 12: On-Demand Self-Service Explained

Heading: On-Demand Self-Service

Subheading: No Human Intervention

Content:

On-demand self-service lets a customer provision computing resources automatically, without requiring a phone call or ticket to the provider's staff.

- Choose the resources you need using a cloud console, command, or API
- No need to wait for IT staff or lengthy hardware-procurement processes
- For example, AWS lets you launch a virtual server (EC2) with a few clicks

Analogy: An ATM lets a customer withdraw cash any time, without a bank clerk approving each transaction — just as on-demand self-service lets a customer provision cloud resources instantly without provider staff involvement.

Visual Guidance: Screenshot-style mockup: AWS console 'Launch Instance' button highlighted, arrow showing instant provisioning with no approval step in between.

Speaker Notes: "Contrast this with traditional data-centre provisioning, which needed weeks of approval tickets. On-demand self-service removes that human approval step entirely."

Slide 13: Broad Network Access

Heading: Broad Network Access

Subheading: Ubiquitous Standard Access

Content:

Broad network access means cloud resources are reachable over standard network protocols from heterogeneous client devices, laptops, phones or thin clients alike.

- Uses standard protocols: HTTPS APIs, SDKs, browser console
- Same service reachable from laptop, mobile app, CLI
- No proprietary client software required, unlike legacy systems
- AWS: identical API reachable via Console, CLI, SDK, mobile app

Visual Guidance: Diagram: single AWS icon at centre, arrows fanning out to laptop, phone, browser and CLI-terminal icons, each arrow labelled 'HTTPS/API'.

Speaker Notes: "This leans on already-known HTTP and client-server concepts, now applied to cloud access. A private, LAN-only application lacks this same universal reachability."

Slide 14: Resource Pooling & Multi-Tenancy

Heading: Pooling And Multi-Tenancy

Subheading: Shared Infrastructure Model

Content:

Resource pooling means provider infrastructure is shared across customers, with capacity assigned dynamically rather than dedicated per customer.

- Multi-tenancy: multiple customers share same physical hardware
- Hypervisor: virtualization software splitting one server into isolated tenants
- Location independence: customer unaware of physical resource
- AWS: EC2 hosts run multiple tenants via shared hypervisor

Analogy: A hostel mess kitchen serves many students from one shared stock of vessels, portioning food on demand instead of giving each student a dedicated stove — exactly as resource pooling shares infrastructure dynamically across tenants.

Visual Guidance: Diagram: one physical server icon with a hypervisor layer, three tenant boxes (A, B, C) each drawing a slice dynamically from the shared pool.

Speaker Notes: "Emphasise that hypervisor-enforced isolation is what keeps tenant data separate on shared hardware. This isolation is what makes multi-tenancy safe despite shared physical infrastructure."

---------- SECTION 4: 4. Essential Characteristics - II----------

Slide 15: Elasticity Scalability And Measured Service

Heading: Three Characteristics Overview

Subheading: Elasticity Scalability Metering

Content:

Cloud platforms must absorb changing demand and charge only for what gets used. Three characteristics make that possible: rapid elasticity, scalability, and measured service.

- Rapid elasticity: automatic scale-out and scale-in with demand
- Scalability: capacity to grow, vertically or horizontally
- Measured service: usage tracked and billed per resource consumed

Visual Guidance: Three-panel diagram: an elastic demand curve with scaling arrows in and out, beside a meter icon feeding a billing screen.

Speaker Notes: "Distinguish elasticity's automatic, reversible scaling from scalability's broader capacity question before tracing a live example. Interviewers often ask for a precise definition of each term, so anchor both here."

Slide 16: Rapid Elasticity Explained

Heading: Rapid Elasticity

Subheading: Auto-Scaling Mechanism

Content:

Rapid elasticity is the cloud's ability to provision and release resources - in most cases automatically, almost instantly, as demand rises or falls.

- Triggered by real-time metrics, such as CloudWatch alarms
- Auto Scaling Groups add or remove instances within minutes
- Scales in both directions: out under load, in after
- Cost: extra capacity billed only while it actually runs

Analogy: A rubber band stretches under load and springs back once released — just as auto-scaled capacity expands during a spike and contracts once demand fades.

Visual Guidance: Timeline graph showing traffic spike with instance-count line rising then falling back to baseline, an Auto Scaling Group icon triggering the change.

Speaker Notes: "Elasticity is automatic and reversible, unlike a one-time capacity upgrade. Interviewers often ask how elasticity differs from simply adding more servers."

Slide 17: Elasticity Vs Scalability Compared

Heading: Elasticity Vs Scalability

Subheading: Traffic Spike Traced

Content:

During a demand surge, elasticity reacts automatically to the moment; scalability asks whether the underlying architecture can absorb growth at all.

Aspect	Elasticity	Scalability
Trigger	Real-time demand (CloudWatch alarm)	Planned capacity growth
Timescale	Minutes, fully automatic	Hours to days, manual or automated
Direction	Scales out and back in	Mostly one-directional growth
Worked case	2→10 instances in ~3 min, back to 2	m5.large resized to m5.4xlarge in advance

Visual Guidance: Side-by-side bar chart: instance count over time showing elasticity's spike-and-return versus a step-change resize for scalability.

Speaker Notes: "Trace the same flash-sale scenario for both terms so the contrast stays concrete. This exact distinction is a frequent interview question, and the trigger and timescale columns are the quick answer."

Slide 18: Measured Service And Metering

Heading: Measured Service

Subheading: Pay-Per-Use Metering

Content:

Measured service means cloud providers automatically monitor, meter and report resource usage, so bills map directly to actual consumption.

- Metrics tracked: compute-seconds, storage-GB, data transfer, API calls
- CloudWatch collects usage; Billing and Cost Explorer compute charges
- Granular units: EC2 per-second, S3 per-GB-month, Lambda per-request
- No usage recorded means no charge — true pay-per-use pricing

Analogy: A household electricity meter records exactly how many units are consumed, and the bill reflects only that usage — just as cloud providers meter every compute-second and byte served, billing only for actual use.

Visual Guidance: Meter icon feeding into a billing dashboard graphic, with small tags showing per-second, per-GB and per-request units.

Speaker Notes: "Metering makes elastic scaling financially viable, since every added instance bills only while running. Interviews often ask how pay-per-use differs from fixed licensing."

---------- SECTION 5: 5. Benefits & Challenges ----------

Slide 19: Cloud Benefits And Challenges Overview

Heading: Benefits & Challenges

Subheading: Cloud Trade-Offs Map

Content:

Cloud adoption delivers clear business benefits, but each benefit carries a matching operational risk; recognising both before migration separates sound architecture decisions from costly surprises.

Benefits	Challenges
Cost model shift (CapEx→OpEx)	Vendor lock-in
Elasticity of resources	Cost overruns
Global reach	Latency
Speed to market	Data residency

Visual Guidance: Two-column split slide: benefits listed left, challenges listed right, with arrows linking related pairs across the middle.

Speaker Notes: "Present benefits and challenges as one connected map rather than two separate lists. Interviewers often ask candidates to name both sides of a cloud trade-off in the same answer."

Slide 20: Cost Model Shift From CapEx To OpEx

Heading: Cost Model Shift

Subheading: CapEx To OpEx

Content:

On-premise IT demands large upfront capital expenditure on hardware before any workload runs. Cloud shifts most of that spend to metered operating expenditure, billed for resources as they're consumed.

- No upfront hardware: Avoid large hardware purchases and depreciation costs.
- Usage-based spending: Costs increase or decrease with resource usage.
- Reserved Instances: Commit to usage for a lower price, with less flexibility.
- Business benefit: Free up capital for other business needs.

- Cost tracking: AWS Cost Explorer and Billing Dashboard help monitor spending.

Analogy: Renting a taxi for occasional trips avoids buying and maintaining a car that sits idle most days — exactly as OpEx cloud billing avoids owning hardware capacity that sits unused.

Visual Guidance: Two bars side by side: a tall one-time CapEx spike versus a flat recurring OpEx line over time.

Speaker Notes: "The shift is financial, not technical — the underlying compute still runs somewhere. A common interview question asks to explain CapEx versus OpEx in cloud adoption."

Slide 21: Elasticity, Global Reach And Speed To Market

Heading: Growth And Speed Benefits

Subheading: Scale Reach Speed

Content:

Elasticity, global reach and speed to market convert cloud's technical flexibility into concrete business value: faster launches, worldwide presence and lower idle spend.

- Elasticity: Automatically scale resources up or down as demand changes.
- Global reach: Deploy applications across AWS Regions worldwide.
- Speed to market: Launch applications and services faster.
- Business value: Scale easily, reach more users, and reduce idle costs.
- Real-world: AWS Auto Scaling groups, Global Infrastructure Region map

Visual Guidance: World map with AWS Region markers lighting up one after another, alongside a rising-then-falling autoscaling curve.

Speaker Notes: "Tie the benefit back to the underlying elasticity mechanism rather than the marketing phrase alone. Interviewers may ask how elasticity concretely shortens time-to-market."

Slide 22: Vendor Lock-In And Switching Costs

Heading: Vendor Lock-In

Subheading: Switching Cost Risk

Content:

Deep reliance on one provider's proprietary APIs, data formats or managed services makes moving a workload elsewhere slow, risky and expensive.

- Proprietary services : Services like Lambda and DynamoDB can make applications harder to move.
- Data egress fees: Moving data out of a provider can add extra costs.
- Specialized skills: Teams may become dependent on one provider's tools.
- Mitigation: Use multi-cloud, open standards, and Infrastructure as Code (IaC)

Analogy: A rented apartment fitted with the landlord's custom cupboards works only in that apartment — just as an application built entirely on one cloud's proprietary APIs runs only on that cloud.

Visual Guidance: Icon of an application boxed inside one provider's logo, with a locked padlock on the arrow pointing toward other providers.

Speaker Notes: "Lock-in is a design-time risk that compounds with every proprietary service adopted, not a one-time migration cost. Interviewers often probe how open standards or multi-cloud design mitigate it."

Slide 23: Cost Overruns In Pay-Per-Use Billing

Heading: Cost Overruns

Subheading: Unplanned Billing Risk

Content:

Pay-per-use billing grows silently with usage, unlike a fixed on-premise budget; unmonitored or misconfigured resources can turn into a large, unexpected bill.

- Unused resources: Resources left running continue to generate charges.
- Autoscaling: Incorrect settings can increase costs unexpectedly.
- Extra charges: Data transfer and API usage can add up quickly.
- Cost control: AWS Budgets and Cost Anomaly Detection help identify unusual spending.

Analogy: Leaving international roaming switched on during travel racks up charges unnoticed until the bill arrives — just as unmonitored cloud resources keep billing until the invoice reveals the overrun.

Visual Guidance: A steadily climbing bill graph with a red spike where an unmonitored resource kept running.

Speaker Notes: "Stress that cost overruns are an operational discipline problem, not a pricing flaw. A frequent interview question asks how to prevent cloud bill shock in production."

Slide 24: Latency And Data Residency Challenges

Heading: Latency And Residency

Subheading: Distance And Law Limits

Content:

Physical distance to the nearest AWS Region adds network delay, and certain data must legally remain within national borders regardless of which Region is closest.

- Latency: Choose a nearby AWS Region or CDN edge location to reduce network delay.
- Data residency: Laws and regulations may require certain data to stay within a country.
- Region selection: Choose the appropriate AWS Region to meet data residency requirements.

Visual Guidance: World map showing a user, a distant Region (long delay arrow) and a nearby Region chosen instead, with a border outline marking a data-residency zone.

Speaker Notes: "Distinguish latency as a performance concern from data residency as a legal constraint, though both push toward the same Region-selection decision. Interviewers may ask why a global application still needs region-specific deployments."

Slide 25: Pairing Cloud Benefits With Their Risks

Heading: Benefit Risk Pairing

Subheading: Weighing Trade-Offs Together

Content:

Each cloud benefit carries a paired risk; weighing both together, not benefits alone, is what a sound migration decision requires.

Benefit Driving Adoption	Matching Risk To Manage
Cost model shift (OpEx)	Cost overruns without governance
Elasticity & speed to market	Vendor lock-in via managed services
Global reach	Latency & data residency limits

Visual Guidance: Balance-scale graphic with a benefit icon on one side and its matching risk icon on the other, for each of the three rows.

Speaker Notes: "Ask which risk applies to which benefit, since exam and interview questions often pair them directly. This pairing is the practical takeaway before any cloud migration decision."

Key Takeaways

1. Cloud Computing Basics: Definition & the NIST view; on-premise IT vs cloud; drivers: cost, scale & speed
2. History & Evolution: Mainframes & time-sharing; client-server & virtualization; grid & utility computing; the public cloud era
3. Essential Characteristics - I: On-demand self-service; broad network access; resource pooling & multi-tenancy
4. Essential Characteristics - II: Rapid elasticity vs scalability; measured service & pay-per-use metering
5. Benefits & Challenges: Cost model shift, elasticity, global reach & speed to market; vendor lock-in, cost overruns, latency & data residency

Upcoming Session : Deployment & Service Models

Thank You | All the Best